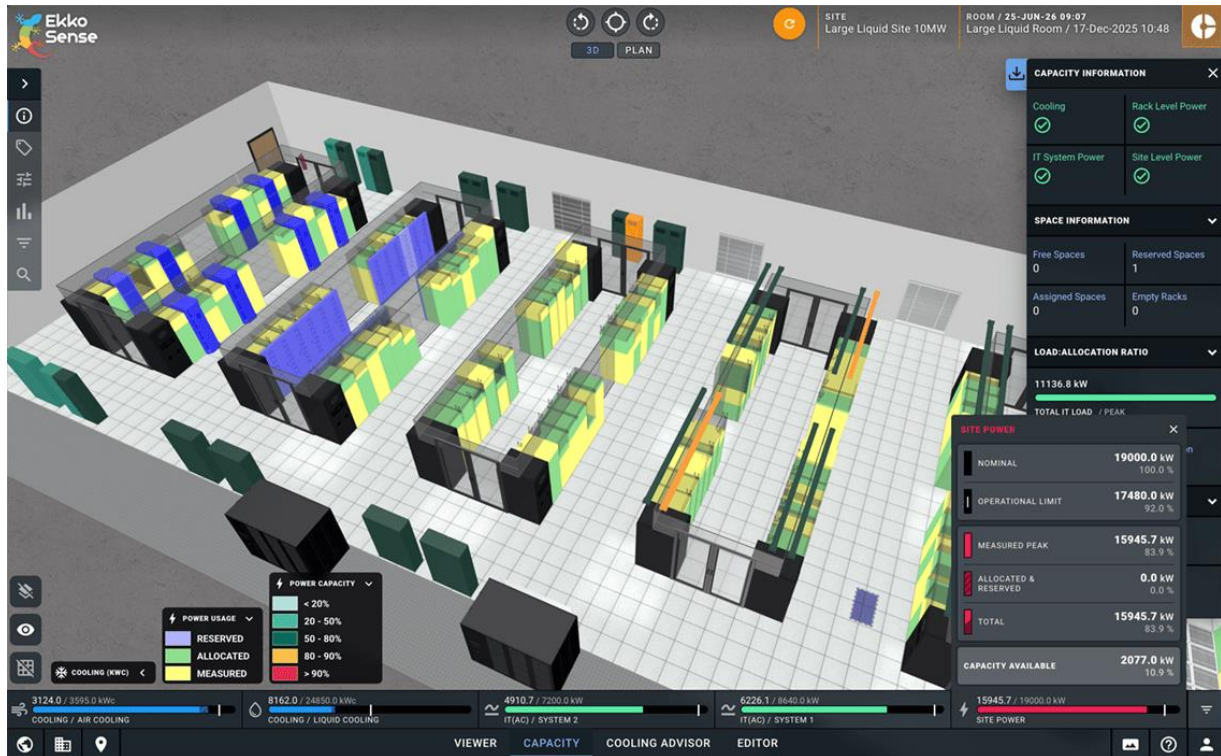


How to find stranded capacity in data centers

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July 17, 2026



Key takeaways: How to find stranded capacity in data centers

- Stranded capacity is power or cooling capacity that exists but remains unusable due to poor visibility into actual infrastructure performance and future planned deployments
- Most data centers operate at 60-70% of design capacity, with 30-40% stranded behind conservative assumptions and monitoring gaps.
- EkkoSense helps data center teams unlock stranded M&E capacity through real-time 3D visualization and AI-driven analytics.

- Recovering stranded capacity offers a faster, more cost-effective alternative to new construction for meeting AI workload demands.
- Continuous monitoring at granular levels transforms capacity planning from periodic audits into ongoing operational intelligence.

What is stranded capacity in data centers?

Stranded capacity refers to power, cooling, or space that is technically available but cannot be safely used because operators lack the real-time and historical matched with forecasted data to validate it. This gap between designed capacity and usable capacity represents one of the most significant hidden challenges facing data center operations.

When an engineer designs a facility rated at 10 MW, conservative safety margins, thermal constraints, and years of derating assumptions can reduce actual usable capacity to 7 MW. That 3 MW difference is stranded – not because the infrastructure failed, but because nobody has the measurement granularity to prove it is safe to use.

The Uptime Institute reports that facilities frequently declare 100% capacity while actual post-optimisation analysis reveals 20-25% of additional available capacity. This stranded capacity sits idle, generating zero return while operators build new facilities to meet demand.

Why does stranded capacity occur?

Conservative Engineering Assumptions Stack Over Time.

Data center infrastructure is designed with safety margins. Operations teams add derating for observed issues. Thermal constraints reduce power below ratings. Equipment ages. Equipment and loads are not laid out as per the original designs. With each passing year, conservative assumptions layer on top of each other, widening the gap between design capacity and what teams feel confident deploying.

Monitoring Blind Spots Hide Available Headroom

Building management systems track consumption – they tell you how much power you are using. They do not tell you how much power you can safely deliver. Traditional monitoring approaches based on interval sampling miss the transient

conditions and equipment health indicators that validate true capacity.

Without granular, real-time and historical telemetry at the full stack (incomer to rack), operations teams do the only logical thing: they overcool, over-provision, and leave capacity on the table to protect against risks they cannot see.

Cooling Capacity Limitations Constrain Usable Power

Power availability means nothing if you cannot cool the equipment it supports. Heat reduces computing performance, damages equipment, and causes outages. Many facilities have power capacity that exceeds their estimated practical cooling capacity, creating thermal constraints that strand megawatts of electrical headroom.

Disconnected Data Sources Prevent Holistic Visibility

Power data lives in one system. Cooling data lives in another. Asset information sits in spreadsheets. When these data sources remain siloed, correlating power availability with thermal conditions and equipment status becomes manual, time-consuming, and error-prone – if it happens at all.

How to identify stranded capacity in your facilities

Map Your Power Chain from Utility to Rack

Start by documenting every component in your power distribution hierarchy: transformers, switchgear, UPS systems, PDUs, and panel boards. Compare nameplate ratings against actual measured loads at each point. Where you find gaps between rated capacity and measured consumption, you have potential stranded capacity.

Pay particular attention to branch circuit utilization. Power monitoring at the circuit level reveals where capacity remains underused due to load imbalances or conservative provisioning policies.

Assess Cooling Capacity Against Actual Thermal Load

Compare your CRAC/CRAH unit output against the actual heat load generated by IT equipment. Temperature monitoring at rack inlets reveals whether cooling is being delivered effectively or wasted through bypass airflow and hotspots.

ASHRAE TC9.9 guidelines indicate that modern IT equipment operates safely with supply temperatures up to 27°C (80.6°F). Facilities running at 18-20°C set points are likely overcooling, wasting energy and potentially masking issues that lead to stranded cooling capacity that could support additional load.

Evaluate Phase Balancing Across Three-Phase Circuits

Unbalanced three-phase loading reduces available capacity below the theoretical maximum. When phases carry unequal loads, the most heavily loaded phase determines the safe operating limit—stranding capacity on the underutilized phases.

Visualizing 3-phase loadings on PDUs helps identify imbalances before they constrain capacity. Addressing these imbalances can recover significant capacity without any infrastructure changes.

Correlate Power, Cooling, and Space Metrics in Real Time

Stranded capacity rarely exists in isolation. A rack may have power headroom but no cooling capacity. A row may have cooling capacity but no available power circuits. Identifying recoverable capacity requires correlating all three dimensions—power, cooling, and space—simultaneously.

EkkoSense's [EkkoSoft Critical platform](#) brings M&E capacity data together in a single 3D visualization, enabling teams to see exactly where constraints exist, what is coming in the future and where capacity can be recovered.

How AI-driven power monitoring uncovers hidden capacity

Real-Time Granular Visibility Replaces Assumptions

Traditional monitoring captures data at intervals – every 5 minutes, 15 minutes, or longer. This sampling approach, mixed with a lack of ability to easily trend historical performance misses transient conditions, peak loads, and equipment

anomalies that determine true safe operating limits. AI-driven monitoring platforms capture data continuously at granular resolution, providing the measurement foundation to validate capacity claims.

When you know the actual thermal state of every rack in near real-time, you can strip away overcooling safety buffers without risking SLAs. The capacity that was stranded behind uncertainty becomes recoverable capacity backed by data.

Machine Learning Identifies Patterns Human Analysis Misses

AI algorithms analyze millions of data points across power, cooling, and environmental sensors to identify correlations and anomalies that would be invisible to manual review. These insights reveal where equipment is trending toward constraints, where headroom exists, and where optimisation opportunities lie.

EkkoSense's Cooling Advisor applies machine learning to over 50 million data points, drawing on deep cooling optimisation expertise to deliver prescriptive recommendations that help teams achieve up to 30% cooling energy savings while releasing stranded capacity.

Continuous Validation Keeps Capacity Data Current

Stranded capacity recovery is not a one-time audit. Equipment conditions change. Load profiles shift. What represents safe headroom today may not be safe headroom in six months. Continuous monitoring transforms capacity planning from a periodic engineering study into a living operational capability. Operations teams see current recoverable capacity rather than stale snapshots from the last manual assessment.

The business case for recovering stranded capacity

Recovered Capacity Delivers Faster Than upgrades or New Construction

Building new data center capacity typically requires 18-36 months and costs between \$7M-\$25M per MW. Permitting, supply chain constraints, and construction delays add risk and extend timelines. Validated capacity recovery, by contrast, can deliver results in 30-90 days at a fraction of the cost.

For operators facing urgent GPU and AI workload demands, the megawatts you already own represent the fastest path to more compute. Recovering stranded capacity buys time while longer-term expansion projects progress.

Revenue Recovery From Unused Power Is Substantial

Stranded capacity generates zero return. At average power rates, each megawatt of recovered and monetised capacity represents roughly \$1-1.3M in annual revenue potential. A facility with 5 MW of recoverable stranded capacity could be leaving \$5-6.5M per year on the table.

Beyond direct revenue, avoiding unnecessary capital expenditure on new construction preserves capital for strategic investments while existing infrastructure delivers its designed value.

Capacity Recovery Supports Sustainability Commitments

Building new facilities consumes embodied carbon and increases total energy demand. Recovering capacity from existing infrastructure delivers compute without additional construction, supporting corporate sustainability targets and regulatory compliance.

Facilities that operate more efficiently – extracting full value from their designed capacity rather than building redundant infrastructure – demonstrate responsible resource stewardship that increasingly matters to customers, investors, and regulators.

Steps to release stranded capacity in your data center

Step 1: Establish a Measurement Baseline

Before optimization begins, establish a 30-90 day baseline measurement period with clearly documented methodology. This baseline becomes your benchmark for demonstrating improvement and calculating ROI. Without credible baseline data, capacity recovery claims lack the evidence to justify operational changes.

Step 2: Deploy Granular Monitoring Across Power and Cooling

Instrument your power chain and cooling systems with sensors capable of providing real-time, continuous data. Focus initial deployment on areas where you suspect stranded capacity: underutilised circuits, rooms running at low thermal density, equipment with significant gaps between nameplate and measured load.

EkkoSense's approach combines low-cost wireless IoT sensors with AI-powered analytics to deliver rack-level visibility without the infrastructure overhead of traditional monitoring deployments.

Step 3: Identify and Validate Recoverable Headroom

Use your monitoring data to identify where capacity exists but remains unused. Validate that this headroom is safe to recover by assessing equipment health, thermal constraints, and operational dependencies. Recoverable capacity is capacity backed by measurement—not capacity assumed from nameplate ratings.

Step 4: Implement Controlled Optimisation

Release stranded capacity incrementally, monitoring the impact of each change. Raise cooling set points gradually while tracking thermal performance at the rack level. Balance phase loading to unlock circuit capacity. Use workflow controls to accept or reject capacity changes based on operational confidence.

Step 5: Maintain Continuous Monitoring and Adjustment

Capacity conditions change continuously. Equipment degrades. Load profiles shift. Maintain ongoing monitoring to ensure recovered capacity remains valid and to identify new optimisation opportunities as they emerge.

How EkkoSense helps data centre teams unlock stranded capacity

Real-Time 3D Visualisation Reveals Capacity Constraints

EkkoSoft Critical provides intuitive 3D visualizations that show exactly where power, cooling, and space constraints exist across your data center estate. Rather than correlating data from disconnected spreadsheets, operations teams see unified views of M&E capacity that enable immediate decisions.

AI-Driven Analytics Quantify Recoverable Capacity

The platform's AI algorithms analyze environmental, power, and cooling telemetry to identify stranded capacity and quantify how much can be safely recovered. With accept/reject workflow controls, teams maintain control over which capacity restrictions to lift based on validated data.

Support for Hybrid Air and Liquid Cooling Environments

As AI workloads drive adoption of liquid cooling alongside traditional air cooling, capacity planning becomes more challenging. EkkoSoft Critical tracks cooling activities and capacity whether you are using air-cooling, liquid cooling, or a hybrid of both—essential visibility for facilities deploying high-density AI compute.

Enterprise-Wide Visibility Across Distributed Estates

For organisations managing multiple facilities, consolidated enterprise dashboards monitor potential capacity issues wherever they occur. This estate-level visibility ensures stranded capacity does not hide in facilities that receive less attention than flagship sites.

What Tools Do You Need to Measure Data Center Infrastructure Efficiency?

Intelligent PDUs/branch circuit monitoring for Granular Power Measurement

Intelligent power distribution units measure power at the outlet or circuit level, providing the granular data foundation for capacity analysis. Look for PDUs that support continuous data collection and integration with your monitoring platform.

Environmental Sensors for Thermal Visibility

Temperature, humidity, and differential pressure sensors reveal where cooling is being delivered effectively and where thermal constraints may be stranding capacity. Deploy sensors at rack inlets and outlets to understand actual thermal conditions rather than relying on room-level averages.

DCIM Platforms for Unified Capacity Management

Next-generation DCIM solutions integrate asset management, power monitoring, and cooling analytics in a single platform. This unified approach eliminates the data silos that hide stranded capacity and enables the correlation analysis required for effective capacity optimization.

AI-Powered Analytics for Continuous Optimization

Machine learning capabilities transform monitoring data into actionable intelligence. AI-driven platforms identify patterns, predict constraints, and recommend optimisation actions that manual analysis would miss or take too long to discover.

FAQs about How to Find Stranded Capacity in Data Centers

What is stranded capacity in a data center?

Stranded capacity is power, cooling, or space that exists within a facility but cannot be used because operators lack the data to validate its safety. EkkoSense helps teams identify and release stranded M&E capacity through real-time 3D visualization and AI-driven analytics that replace conservative assumptions with measured confidence.

How much stranded capacity do typical data centers have?

Most facilities operate at 60-70% of their design capacity, with 30-40% remaining stranded. Uptime Institute research indicates that data centers declaring 100% capacity often have 20-25% additional capacity available post-optimization. EkkoSense customers have recovered significant stranded capacity by deploying granular monitoring and AI-powered analytics.

Why can't traditional BMS/PME tools identify stranded capacity?

Building management systems track power consumption, but they do not validate how much power can be safely delivered, what is allocated and what's coming in the future. Identifying stranded capacity requires understanding actual equipment health, thermal conditions, and operational headroom – data that interval-based BMS

monitoring cannot provide. EkkoSense's continuous, granular approach delivers the visibility that traditional tools miss.

How quickly can stranded capacity be recovered?

Validated capacity recovery typically takes 30-90 days, compared to 18-36 months for new construction. EkkoSense enables data center teams to establish measurement baselines, identify recoverable headroom, and implement controlled optimisation quickly – delivering capacity when it is needed rather than years later.

What are the financial benefits of recovering stranded capacity?

Each megawatt of recovered and monetized capacity represents roughly \$1-1.3M in annual revenue potential. Beyond direct revenue, avoiding unnecessary new construction preserves capital for strategic investments. EkkoSense's software-driven approach delivers typical ROI in under 12 months through cooling energy savings and released capacity.

How does AI improve data center power monitoring?

AI analyzes millions of data points to identify correlations, anomalies, and optimization opportunities that would be invisible to manual review. EkkoSense's Cooling Advisor uses machine learning insights from over 50 million data points to deliver prescriptive thermal advice, helping teams achieve up to 30% cooling energy savings while releasing stranded capacity.

Further research – check out EkkoSoft Critical [data center capacity management](#) capabilities and how it could help your business. Additional video Tech Tip here [“What capacity do you have across your data center estate right now?”](#)